HW #3

1) WSS $x(t)$, $C_{xx}(\tau) = e^{-|\tau|}$, $P_x(f) = \frac{2}{1+(2\pi f)^2}$

A) $P_x(0) = \max = \frac{2}{1+0} = (2)$.

3 dB or 1/2 = 1.

~~$P(f)$~~ $1 = \frac{2}{1+(2\pi f)^2}$ $\quad 1+(2\pi f)^2 = 2$

$(2\pi f)^2 = 1$

$f = \frac{1}{2\pi} = 0.159$ Hz.

B) $\int_{-B}^{B} P_x(f)\,df = 0.9 \cdot \underbrace{\int P(f)\,df}_{\hookrightarrow C_{xx}(0) = 1}$.

$\int \frac{2}{1+(2\pi f)^2}\,df$.

$\frac{1}{\pi} \tan^{-1}(2\pi f) \Big|_{-B}^{B} = 0.9$.

$2\left(\frac{1}{\pi}\tan^{-1}(2\pi B)\right)$ symetry $= 0.9$.

$B = 1.005$ Hz

c) 2x of 3dB $= 2(0.159) = 0.318$ Hz.

$T = \frac{1}{0.318} = 3.145$ s $\rightarrow$ exact is actually $\frac{1}{\frac{1}{2\pi}\cdot 2} = \pi$

π s

D) $C_{xx}[n] = C_{xx}(nT) = e^{-|\pi n|} = e^{-\pi|n|}$

E) $P_x(e^{j2\pi f}) = \sum C_{xx}[k]\, e^{-j2\pi fk}$

$= \sum e^{-\pi|k|}\, e^{-j2\pi fk}$

= use shortcut = $\frac{1-r^2}{1-2r\cos\theta + r^2}$ $\quad r = e^{-\pi}$

$= \frac{1-e^{-2\pi}}{1-2e^{-\pi}\cos(2\pi f) + e^{-2\pi}}$

$\cos(2\pi ft) = \frac{e^{j2\pi ft} + e^{-j2\pi ft}}{2}$ $e^{-j\omega t_0}$

HW #3

2) $f_c = 1700$ MHz ; $W = 20$ MHz.

* $h_c(t) = \sum_{k=0}^{9} (0.8)^k \delta(t - k10^{-7})$ } multipath

A) $H_c(w) = \sum_{k=0}^{9} (0.8)^k \mathcal{F}\{\delta(t - k10^{-7})\}$

$= \sum_{k=0}^{9} (0.8)^k$

$H_c(w) = \sum_{k=0}^{9} (0.8)^k \left(e^{-jwk10^{-7}}\right)$

* $\boxed{H_c(f) = \sum_{k=0}^{9} (0.8)^k \left(e^{-j2\pi fk10^{-7}}\right)}$

↳ In geo form, $a_1\left(\frac{1-r^n}{1-r}\right)$ $a_1 = 1$

$\boxed{1\left(\frac{1-(0.8e^{-j2\pi f10^{-7}})^{10}}{1-0.8e^{-j2\pi f10^{-7}}}\right)}$

$\text{rect}\left(\frac{f}{w}\right) \leftrightarrow W\text{sinc}(W$

B) Define:

$h_p(t) \equiv h_c(t) * p(t)$

$P(f) = \text{rect}\left(\frac{f-f_c}{w}\right) + \text{rect}\left(\frac{f+f_c}{w}\right)$

$p(t) = W\text{sinc}(wt)e^{-j2\pi f_c t} + W\text{sinc}(wt)e^{+j2\pi f_c t}$

$= W\text{sinc}(wt)\left[e^{-j2\pi f_c t} + e^{j2\pi f_c t}\right]$

$= 2W\text{sinc}(wt)\cos(2\pi f_c t)$

$h_p(t) = \left[\sum_{k=0}^{9} (0.8)^k \delta(t - k10^{-7})\right] * 2W\text{sinc}(wt)\cos(2\pi f_c t)$

↓ ↓ ↓ Apply sifting

~~$h_p(t) = \sum_{k=0}^{9} (0.8)^k \delta(t$~~

$\boxed{h_p(t) = \sum_{k=0}^{9} (0.8)^k 2W\text{sinc}(w(t - k10^{-7}))\cos(2\pi f_c(t - k10^{-7}))}$

HW #3 (Cont)

2c) ~~Define~~ $h(t)$ ~~ε~~ Using 2B as starting point

$$h(t) = \sum_{k=0}^{9} 0.8^k e^{-2\pi f_c 10^{-7} k} \text{Wsinc}(W(t - k10^{-7}))$$

2d) $T = \frac{1}{W} = 50\,ns$

$$h[n] = \sum_{k=0}^{9} 0.8^k e^{-j2\pi f_c T_k} \text{sinc}\left(n - \frac{T_k}{T}\right) \qquad T_k = k10^{-7}$$

$$T_k/T = 2k.$$

$$h[n] = \sum_{k=0}^{9} 0.8^k e^{-j2\pi f_c k10^{-7}} \text{sinc}(n - 2k)$$

always integer multiple of 2π

$e^{-j2\pi k}$ for $k \in \mathbb{Z} = \cos(2\pi k) \pm j\sin(2\pi k)$

(1)

$$h[n] = \sum_{k=0}^{9} 0.8^k \text{sinc}(n - 2k)$$

HW #3

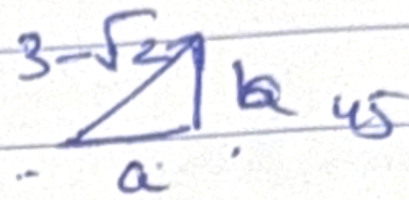
3A) $\mu = \frac{1}{M} \sum s_m$. all points are symmetric around 1 center, which $(1+j)$.

$$\boxed{\mu = 1+j}$$

B) ~~Inner~~

$(3-\sqrt{2}) + 0j$

~~$\det$~~ $3\sqrt{2}$ / 2



45-45 Triangle

$a\sqrt{2} = 3-\sqrt{2}$

$a = \frac{3-\sqrt{2}}{\sqrt{2}}$

$\frac{3}{\sqrt{2}} - 1$

$\frac{3\sqrt{2}}{2} - 1$

Constellation (inner):

$(3-\sqrt{2}) + 0j$

$3\sqrt{2}/2 + 3\sqrt{2}/2\,j$

$0 + (3-\sqrt{2})$

$-3\sqrt{2}/2 + 3\sqrt{2}/2\,j$

$-(3-\sqrt{2}) + 0j$

$-3\sqrt{2}/2 - 3\sqrt{2}/2\,j$

$0 - (3-\sqrt{2})j$

$3\sqrt{2}/2 - 3\sqrt{2}/2\,j$

let $x = \frac{3\sqrt{2}}{2}$

AND

Constellation (outer):

$3 + 0j$

$x + xj$

$0 + 3j$

$-x + xj$

$-3 + 0j$

$-x - xj$

$0 - 3j$

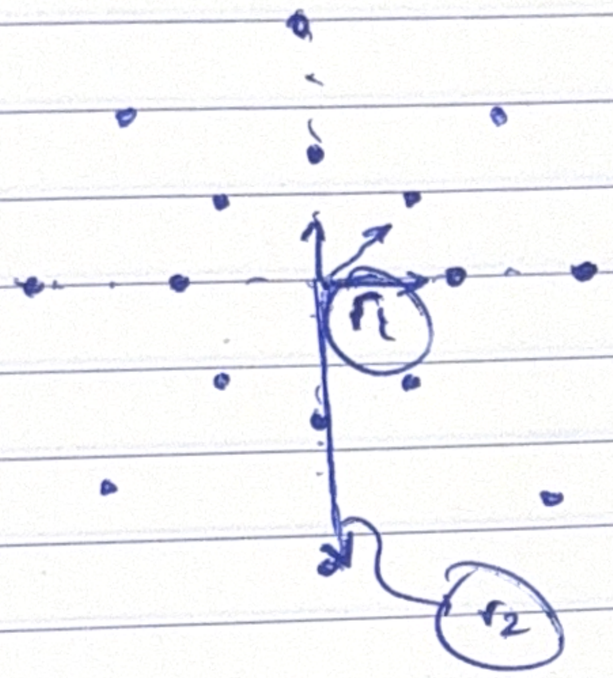
$x - xj$

HW #3 (Cont)

c) $E_{Avg} = (3-\sqrt{2})^2 \cdot \frac{8}{16} + (3)^2 \cdot \frac{8}{16} = \boxed{5.757}$

D) E_{Avg} (unit scaling) $= \frac{1}{\sqrt{E_{Avg}}} \approx \boxed{0.417}$

E)



$r_1 = \frac{(3-\sqrt{2})}{\sqrt{E_{Avg}}} \approx \underline{0.661}$

$r_2 = \frac{(3)}{\sqrt{E_{Avg}}} \approx 1.250$